



# IT582: Foundations of Machine Learning

## Lab-4: Regression using Basis Functions

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### Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.  
The notebook should contain all code, outputs, plots, observations and conclusions.  
Name the notebook as:

[studentid]\_lab4.ipynb

Use the same Linear Regression fitting procedure throughout the lab. Do not use built-in basis-expansion utilities.

### Task 1: Basis Functions and their Support

A basis-function regression model is

$$h_{\theta}(x) = \sum_{j=0}^M \theta_j \phi_j(x), \quad \phi_0(x) = 1.$$

For

$$x \in [-3, 3],$$

construct and plot the following basis functions.

- **Linear:**

$$\phi_1(x) = x.$$

- **Polynomial:**

$$\phi_j(x) = x^j, \quad j = 1, \dots, 5.$$

- **Gaussian:**

$$\phi_j(x) = \exp \left[ -\frac{(x - \mu_j)^2}{2s^2} \right],$$

where

$$\mu_j \in \{-2, -1, 0, 1, 2\}, \quad s = 0.5.$$

- **Sigmoid:**

$$\phi_j(x) = \frac{1}{1 + \exp[-(x - \mu_j)/s]},$$

using the same  $\mu_j$  and  $s = 0.5$ .

Then, for

$$x_0 = 0.3, \quad \delta = 0.1,$$

calculate

$$|\phi_j(x_0 + \delta) - \phi_j(x_0)|$$

for the polynomial, Gaussian and sigmoid bases.

Write observations explaining the **global and local behaviour** of these basis functions.

## Task 2: Polynomial Basis for a Smooth Global Trend

Generate

$$n = 150$$

samples uniformly over

$$x \in [-2, 2]$$

using

$$y = 1 + 0.8x - 1.5x^2 + 0.5x^3 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.2^2).$$

Split the data into 60% training, 20% validation and 20% test sets.

Fit the data using:

- Linear basis
- Polynomial basis of degree 2
- Polynomial basis of degree 3
- Polynomial basis of degree 5

For each model, report Training, Validation and Test MSE.

Plot the data and all fitted curves on the same figure.

Compare the models and write observations on:

- Effect of increasing polynomial degree
- Why polynomial basis functions are suitable for a smooth global trend

## Task 3: Local and Transition-based Regression

### A. Gaussian Basis

Generate

$$n = 150, \quad x \in [-3, 3],$$

using

$$y = 0.5 + 2 \exp \left[ -\frac{(x + 1.2)^2}{2(0.35)^2} \right] - 1.5 \exp \left[ -\frac{(x - 1.0)^2}{2(0.45)^2} \right] + \epsilon,$$

where

$$\epsilon \sim \mathcal{N}(0, 0.1^2).$$

Use 9 equally spaced Gaussian centers over  $[-3, 3]$  and compare

$$s \in \{0.25, 0.5, 1.0\}.$$

Report Validation and Test MSE for each value of  $s$  and plot the fitted curves.

Select the best value of  $s$  and explain why Gaussian basis functions are suitable for this problem.

### B. Sigmoid Basis

Generate

$$n = 150, \quad x \in [-5, 5],$$

using

$$y = 1 + 4 \left[ \frac{1}{1 + \exp(-(x - 0.5)/0.8)} \right] + \epsilon,$$

where

$$\epsilon \sim \mathcal{N}(0, 0.15^2).$$

Use sigmoid basis functions with

$$\mu_j \in \{-2, -1, 0, 1, 2\}$$

and compare

$$s \in \{0.3, 0.8, 1.5\}.$$

Report Validation and Test MSE and plot the fitted curves.

Explain the effect of  $s$  and why sigmoid basis functions are suitable for a saturation or transition-type relationship.

## Task 4: Comparison of Basis Functions

Generate

$$n = 200, \quad x \in [-4, 4],$$

using

$$y = 1 + 2.5 \left[ \frac{1}{1 + \exp(-(x + 0.8)/0.7)} \right] + 1.2 \exp \left[ -\frac{(x - 1.8)^2}{2(0.35)^2} \right] + \epsilon,$$

where

$$\epsilon \sim \mathcal{N}(0, 0.12^2).$$

Use a 60%–20%–20% training, validation and test split.

Fit the data using:

- Linear basis
- Polynomial basis with

$$p \in \{3, 5, 7\}$$

- Gaussian basis with 9 equally spaced centers and

$$s \in \{0.3, 0.6, 1.0\}$$

- Sigmoid basis with 9 equally spaced centers and

$$s \in \{0.3, 0.6, 1.0\}$$

For polynomial, Gaussian and sigmoid models, select the best parameter using Validation MSE.

For the final selected model from each basis family, report:

- Selected degree or width, where applicable
- Training MSE
- Validation MSE
- Test MSE

Plot the fitted curves from all four basis-function families together with the data.

Write a final comparison explaining:

- Which basis gives the best performance and why
- Which basis represents the localized change most naturally
- Which basis represents the broad transition most naturally
- Why one basis may be more suitable than another for a particular structure in the data